Background on Statistics

Bryan Pardo

Northwestern University

(updated fall 2022)

Some basic definitions

Some definitions

- **Sample Space:** The set, S,of possible values a random variable can take. These are mutually exclusive (e.g. coin flips: heads or tails).

- **Random Variable**: A mapping from a sample space to actual measured outputs. We denote the whole mapping with a capital letter (e.g. X) and a particular sample output with a lower case (e.g. x)

- **Support:** For a random variable X, is the portion of the sample space that has non-zero probability. (if a flipped coin always turns up “heads”, then the support is “heads”)

Some definitions

- A **Discrete random variable** has a countable sample space. For us, we'll use FINITE sample spaces, like heads/tails or words in a dictionary.

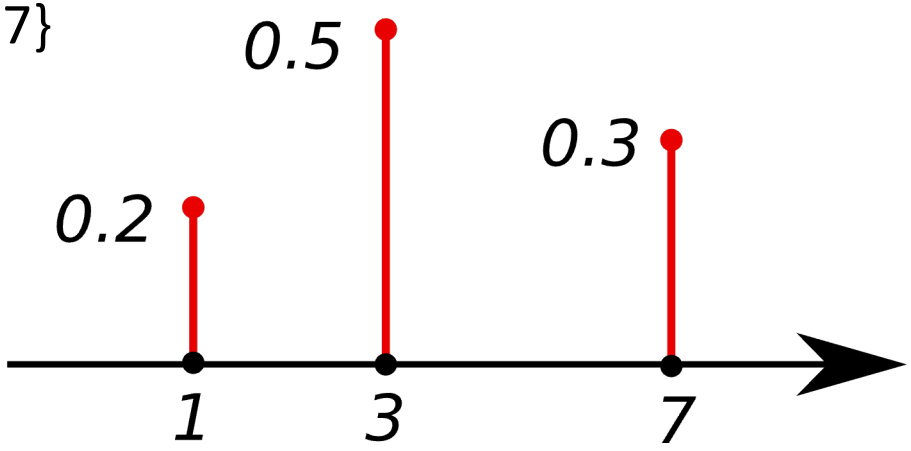
- A **Continuous random variable** has an uncountably infinite sample space, like real numbers on the interval (0,1).

Probability Mass Function

… specifies the probability of a **discrete random variable** taking each of the values in the sample space.

The PMF is nonnegative, and the sum of its probabilities = 1.0

In this example, the sample space is {1,3,7}



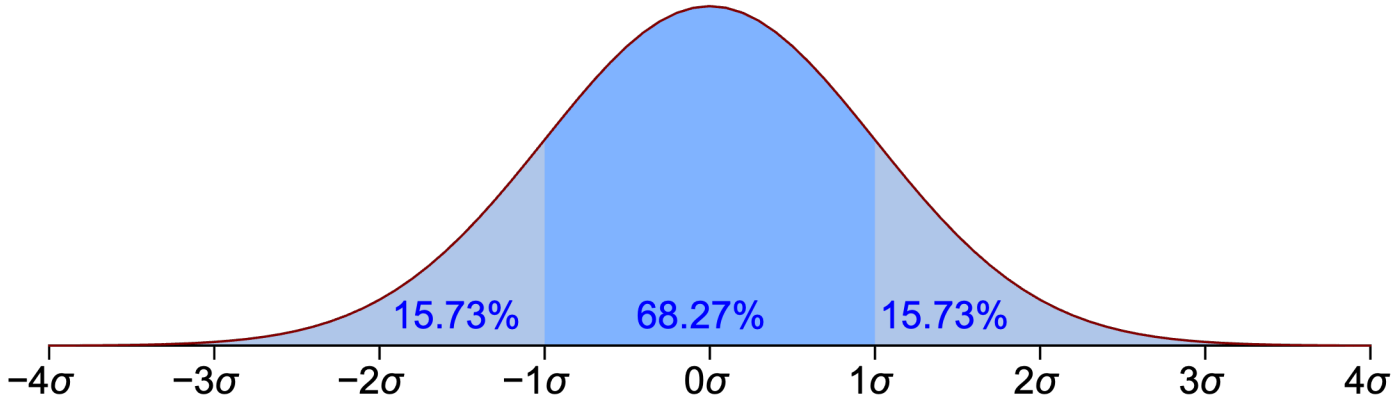
By Oleg Alexandrov - Own work, Public Domain, https://commons.wikimedia.org/w/index.php?curid=2073424

Probability density function (PDF)

… specifies the probability of a **continuous random variable** falling within a particular range of values.

This probability is given by the integral of of the PDF over that range (i.e. the area under the curve).

The PDF is nonnegative, and its integral over the entire space = 1.0



By Jhguch at en.wikipedia, CC BY-SA 2.5,
https://commons.wikimedia.org/w/index.php?curid=14524285

Expected value E[X] of a random variable X

- For a finite, discrete random variable X, E[X] is defined as

$$E[X] = \sum_{x \in S} xp(x)$$

x is a particular element in the sample space S
p(x) is the probability mass function for X applied to the sample x.

Expected value E[X] of a random variable X

- For a finite, discrete random variable X, E[X] is defined as

$$E[X] = \sum_{x \in S} xp(x)$$

This is the probability-weighted average value. For example, let's say that we have a fair 6-sided die. Then every p(x) = 1/6. The expected value of the die is…

$$E[X] = \left(\frac{1}{6}\right)1 + \left(\frac{1}{6}\right)2 + \left(\frac{1}{6}\right)3 + \left(\frac{1}{6}\right)4 + \left(\frac{1}{6}\right)5 + \left(\frac{1}{6}\right)6 = \frac{21}{6} = 3.5$$

Expected value E[X] of a random variable X

- For a continuous random variable on real numbers, E[X] is defined as

We often drop the -∞, ∞ and assume they're implied

$$E[x] = \int_{-\infty}^{\infty} xp(x)dx$$

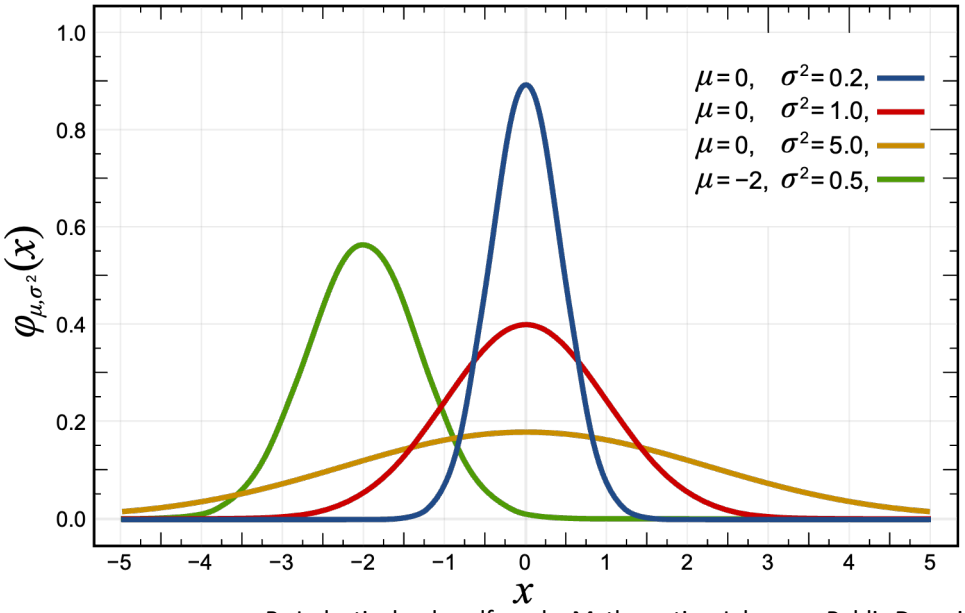
x is a particular element in the sample space S

p(x) is the probability mass (or density) function for X applied to the sample x.

The Normal (Gaussian) Distribution

The Gaussian a.k.a. Normal Distribution

$$x \sim N(\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



By Inductiveload - self-made, Mathematica, Inkscape, Public Domain, https://commons.wikimedia.org/w/index.php?curid=3817954

The multivariate normal distribution of a *k*-dimensional random vector $\mathbf{X} = (X_1, \ldots, X_k)^{\mathrm{T}}$ following notation:

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

or to make it explicitly known that *X* is *k*-dimensional,

$$\mathbf{X} \sim \mathcal{N}_k(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

with *k*-dimensional mean vector

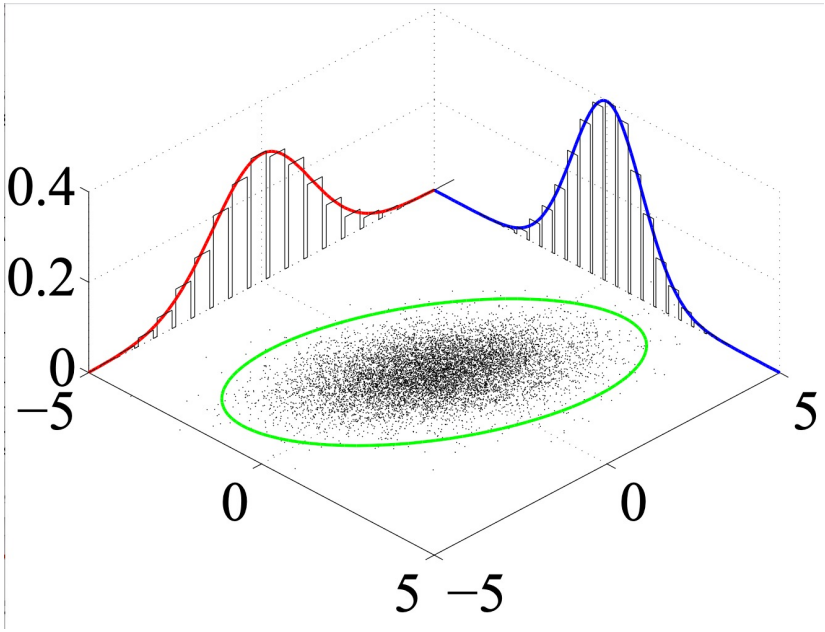
$$\boldsymbol{\mu} = \mathrm{E}[\mathbf{X}] = (\mathrm{E}[X_1], \mathrm{E}[X_2], \ldots, \mathrm{E}[X_k])^{\mathbf{T}},$$

and $k \times k$ covariance matrix

$$\Sigma_{i,j} = \mathrm{E}[(X_i - \mu_i)(X_j - \mu_j)] = \mathrm{Cov}[X_i, X_j]$$

such that $1 \leq i \leq k$ and $1 \leq j \leq k$.

https://en.wikipedia.org/wiki/Multivariate_normal_distribution#Notation_and_parameterization



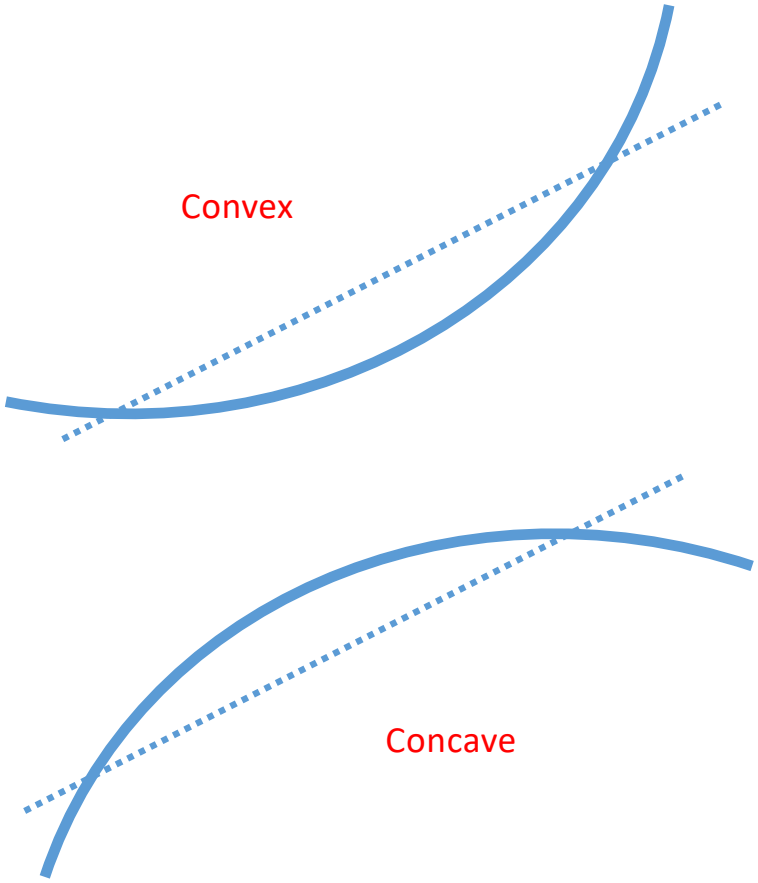
By IkamusumeFan - Own work, CC BY-SA 3.0,
https://commons.wikimedia.org/w/index.php?curid=30432580

Jensen’s Inequality

Some definitions

- A real-valued function is **convex** if the line segment between any two points on the graph of the function lies **above** the graph between the two points.

- A real-valued function is **concave** if the line segment between any two points on the graph of the function lies **below** the graph between the two points.



Is my function convex/concave?

- A differentiable function is (strictly) convex if its second derivative is (strictly) positive;
- That function is (strictly) concave if its second derivative is (strictly) negative.
- Example The natural logarithm is strictly concave because…

$$\frac{d^2}{dx^2}\log(x) = -x^{-2}$$

Now for the inequality…

- let X be an integrable random variable a random variable and ϕ be a concave function, then Jensen's inequality is defined as:

$$E[\phi(X)] \leq \phi(E[X])$$

- We know log is a concave function. This means:

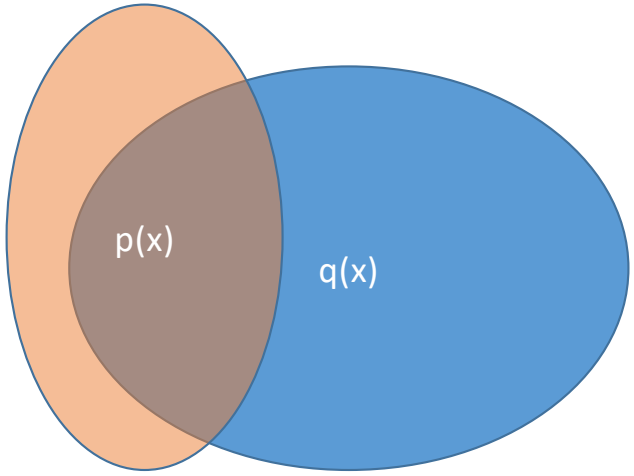
$$E[log(X)] \leq log(E[X])$$

- …we'll be using this later.

Kullback-Leibler divergence

Comparing two PDFs

- Given two PDFs, p(x) and q(x), how do I tell how similar they are?
- Maybe we could measure the overlap in some way?
- People use Kullback-Leibler divergence (KL divergence) for this.



KL-divergence

- P and Q are both probability functions on the same sample space.
- For finite sample spaces:

$$D_{KL}(P||Q) = \sum_{x} P(x) log\left(\frac{P(x)}{Q(x)}\right)$$

- For real-valued sample spaces :

$$D_{KL}(P||Q) = \int p(x) log\left(\frac{p(x)}{q(x)}\right) dx$$

Relation to entropy

$$D_{KL}(P||Q) = \sum_x P(x) log\left(\frac{P(x)}{Q(x)}\right)$$

$$= \sum_x P(x)\log(P(x)) - \sum_x P(x)\log\big(Q(x)\big)$$

This is the entropy of P(x)

This is the cross-entropy of P(x) compared to Q(x)

Relation to expected value

$$D_{KL}(Q(z)||P(z|x)) = \sum_{z} Q(z) log\left(\frac{Q(z)}{P(z|x)}\right)$$

$$= \sum_{z} Q(z)(\log Q(z) - \log P(z|x))$$

$$= E_{z\sim Q}[(\log Q(z) - \log P(z|x))]$$

Jensen-Shannon Divergence

- Symmetric measure
- People often use this, because it is symmetric and avoids divide by 0

$$D_{JS}(P||Q) = D_{KL}(P||M) + D_{KL}(Q||M)$$

…where $M = \frac{P+Q}{2}$

Joint distributions

Joint Distribution

- Given two random variables, X and Z, defined on the same probability space, the joint probability distribution is the probability distribution on all possible pairs of outputs.

$$\mathrm{P}(X, Z)$$

By IkamusumeFan - Own work, CC BY-SA 3.0,
https://commons.wikimedia.org/w/index.php?curid=30432580

The Chain Rule & Conditional Probability

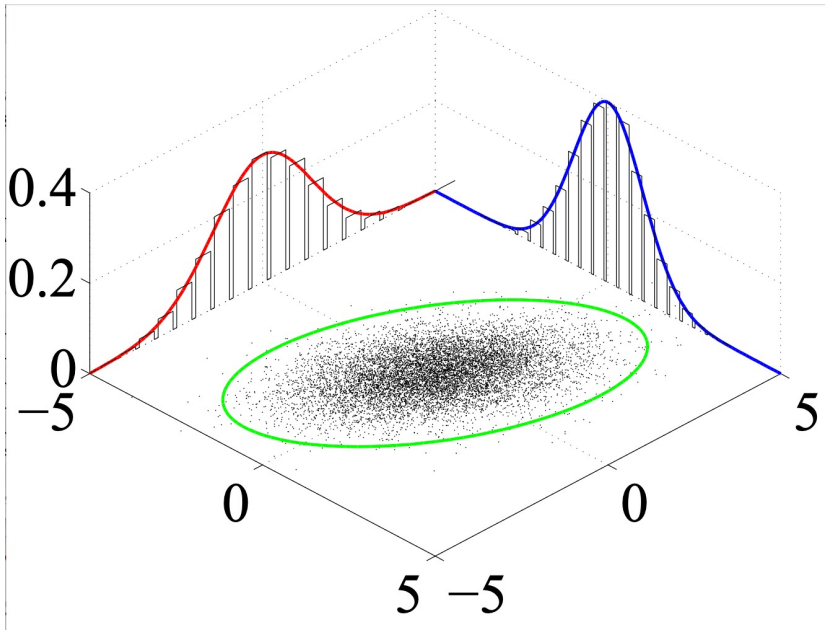
Definition of conditional probability:

$$P(X|Z) = P(X,Z)/P(Z)$$

$$P(Z|X) = P(X,Z)/P(X)$$

We can factor the joint distribution using conditional probability:

$$\begin{aligned} P(X,Z) &= P(X|Z)P(Z) \\ &= P(Z|X)P(X) \end{aligned}$$



By IkamusumeFan - Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=30432580

Bayes Rule

With conditional probability:

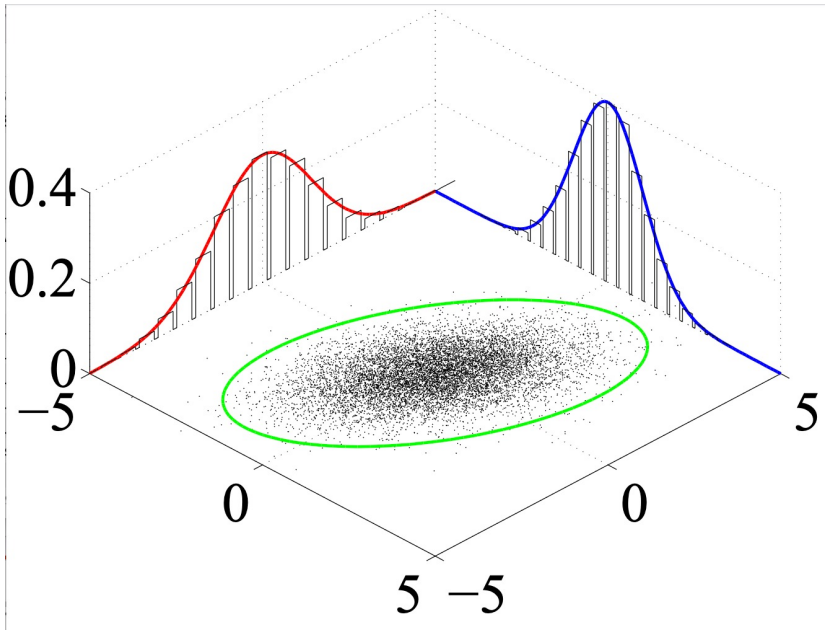
$$P(X|Z) = P(X,Z)/P(Z)$$

…and the chain rule:

$$P(X,Z) = P(Z|X)P(X)$$

We get to Bayes rule:

$$\begin{aligned} P(Z|X) &= P(X,Z)/P(X) \\ &= P(X|Z)P(Z)/P(X) \end{aligned}$$



By IkamusumeFan - Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=30432580

A finite Joint Distribution of Booleans

- A truth table listing all combinations of variable values
- This embodies P(A,B,C)
- Each combination has a probability that must be estimated
- How big is this table for 100 Boolean variables?

A	B	C	Prob
0	0	0	0.1
0	0	1	0.2
0	1	0	0.1
0	1	1	0.05
1	0	0	0.05
1	0	1	0.2
1	1	0	0.25
1	1	1	0.05

Finding a marginal distribution

- To find P(A) we must “marginalize”
- Sum the probabilities of all rows where A=1

$$P(A) = \sum_b \sum_c P(1, b, c)$$

$$= 0.05 + 0.2 + 0.25 + 0.05$$

$$= 0.55$$

What happens if there are 100 variables?

A	B	C	Prob
0	0	0	0.1
0	0	1	0.2
0	1	0	0.1
0	1	1	0.05
1	0	0	0.05
1	0	1	0.2
1	1	0	0.25
1	1	1	0.05